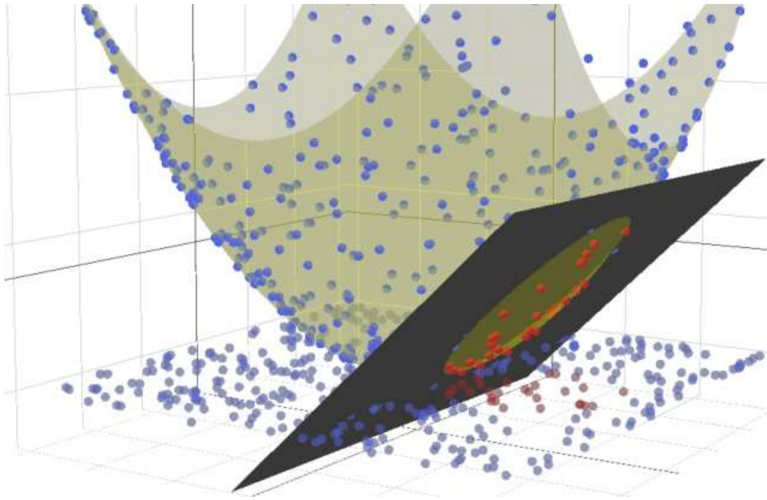
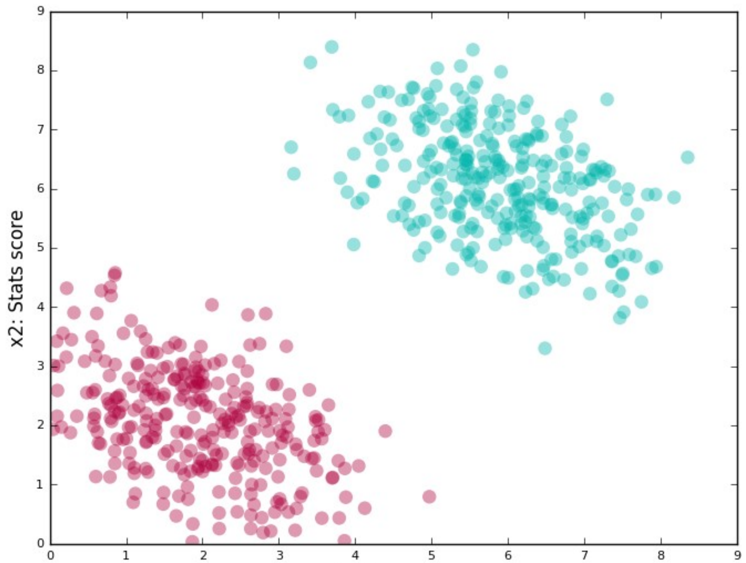
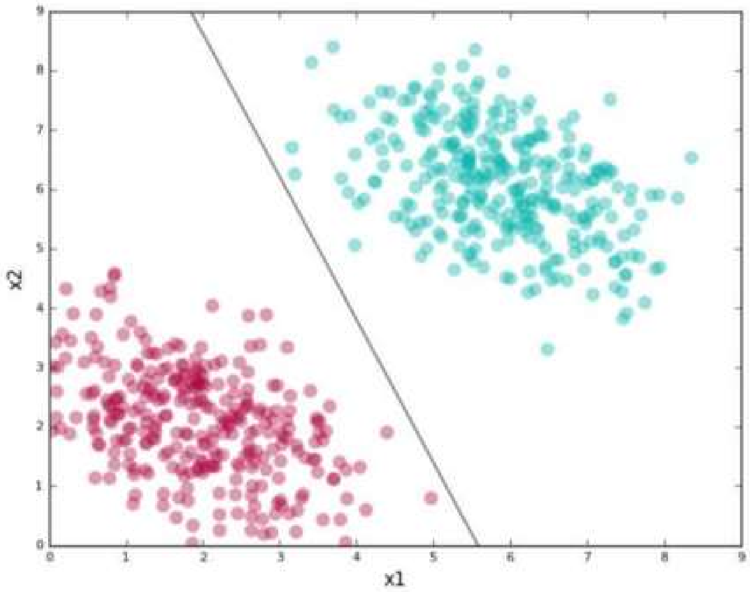
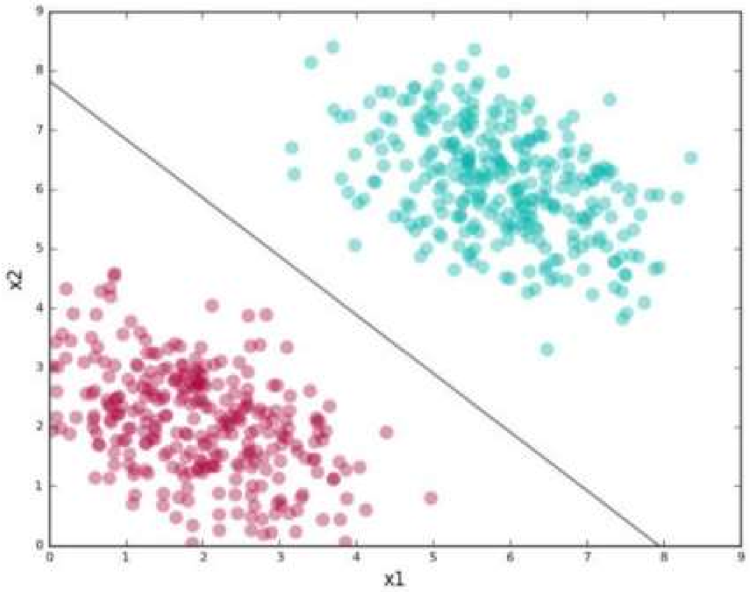
Support Vector Machines

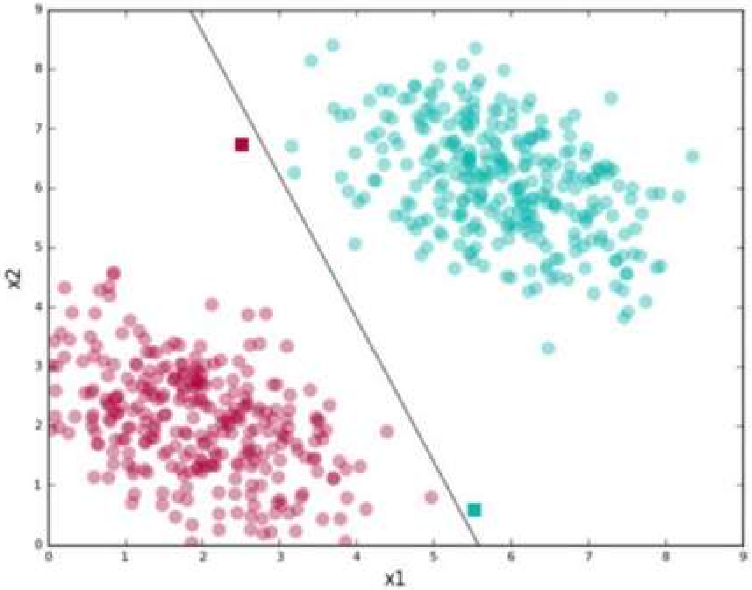
Yongmei Tan
ymtan@bupt.edu.cn

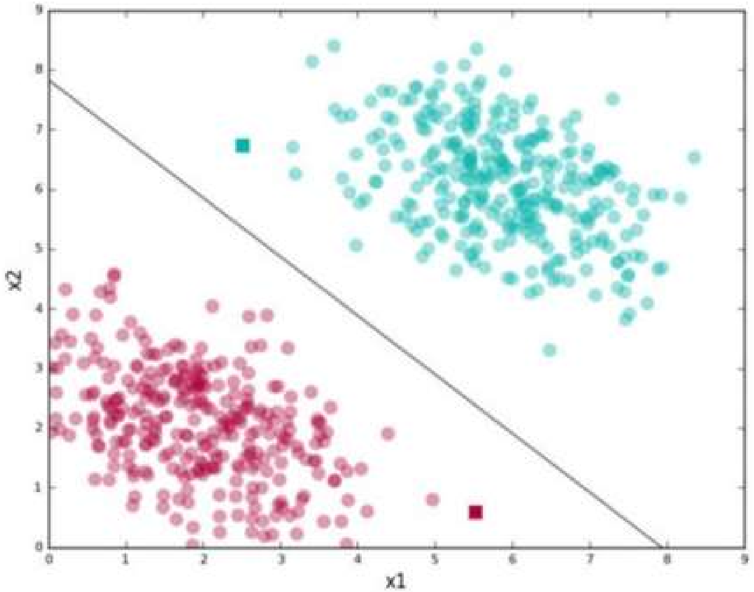


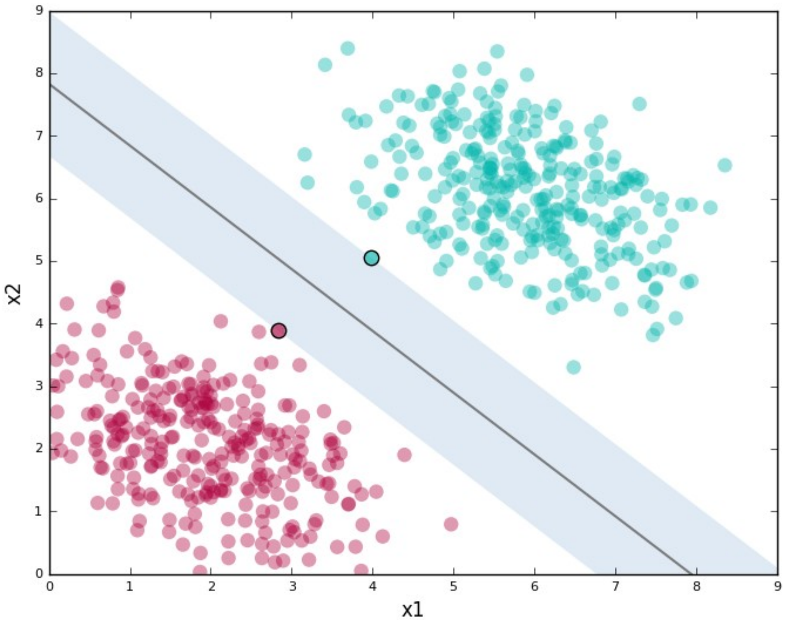

x2: Stats score
0
1
2
3
4
5
6
7
8
9

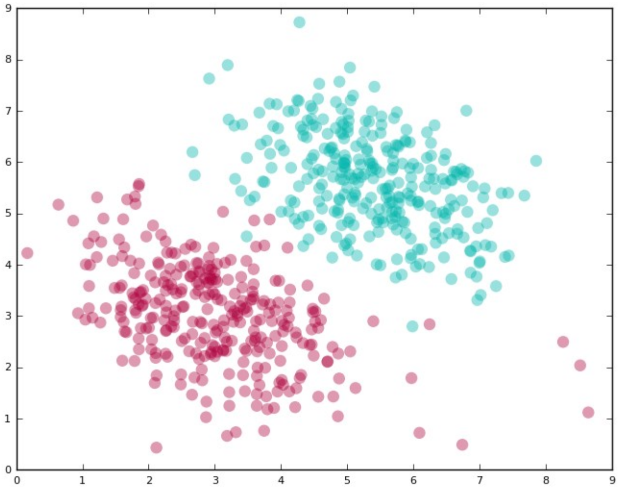

x1
x2
0
1
2
3
4
5
6
7
8
9

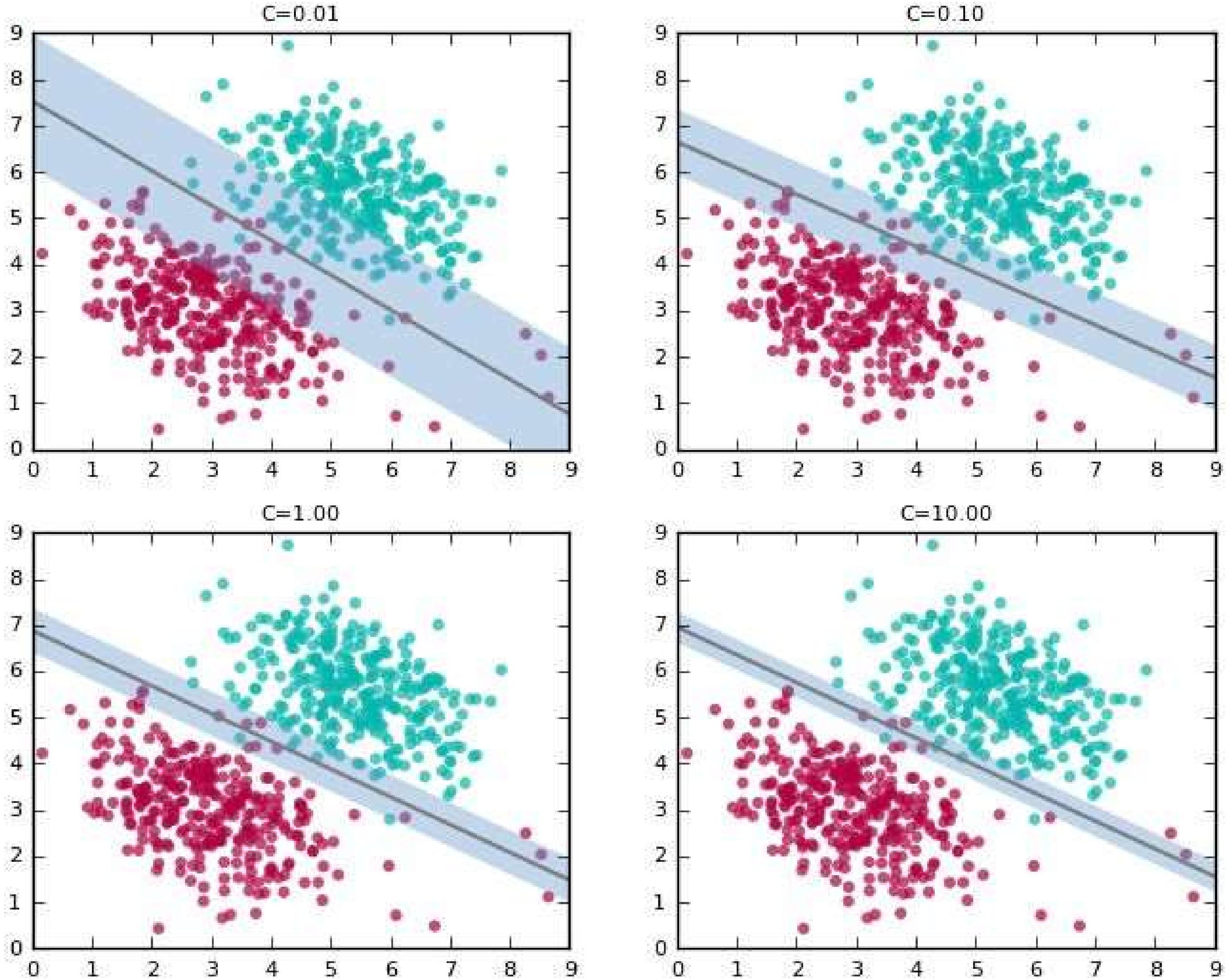

x1
x2
0
1
2
3
4
5
6
7
8
9

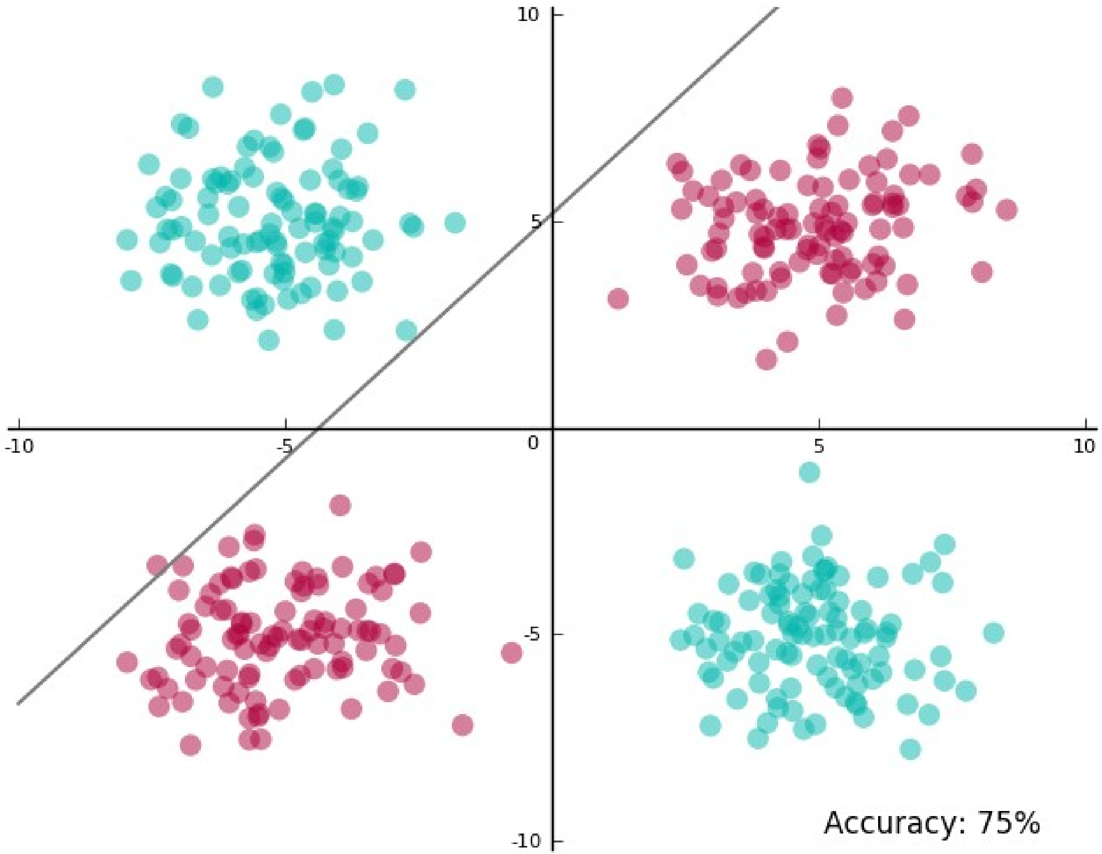

x2
x1
0 1 2 3 4 5 6 7 8 9


x2
x1
0 1 2 3 4 5 6 7 8 9

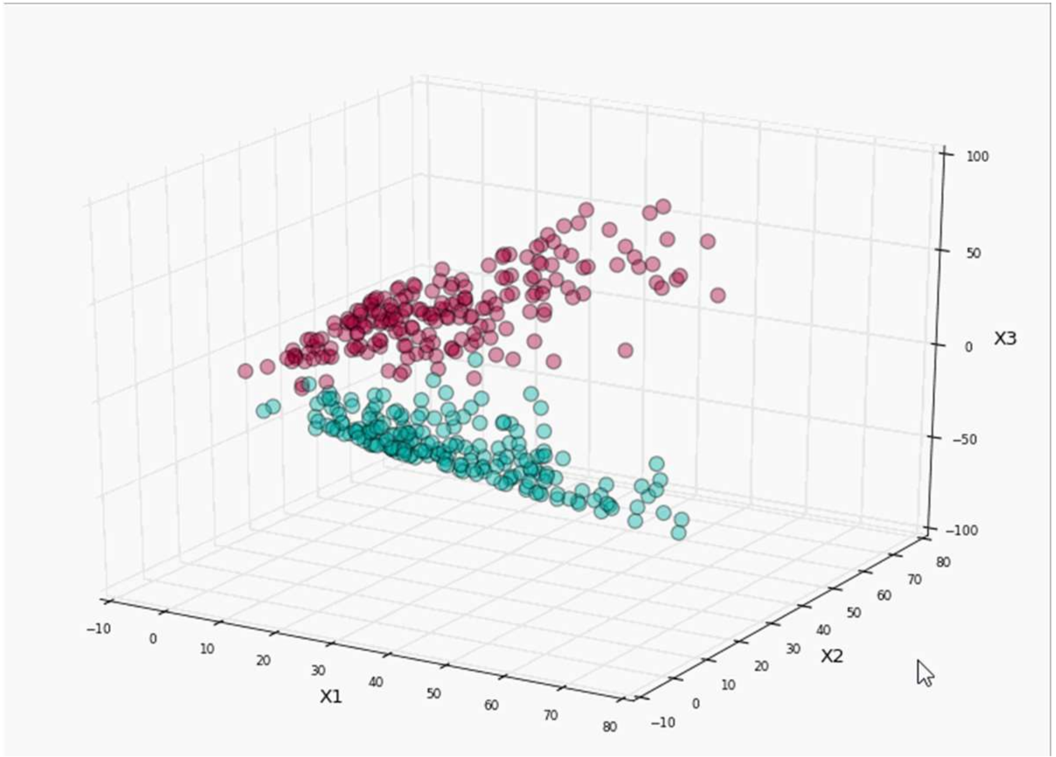

x2
x1
0 1 2 3 4 5 6 7 8 9

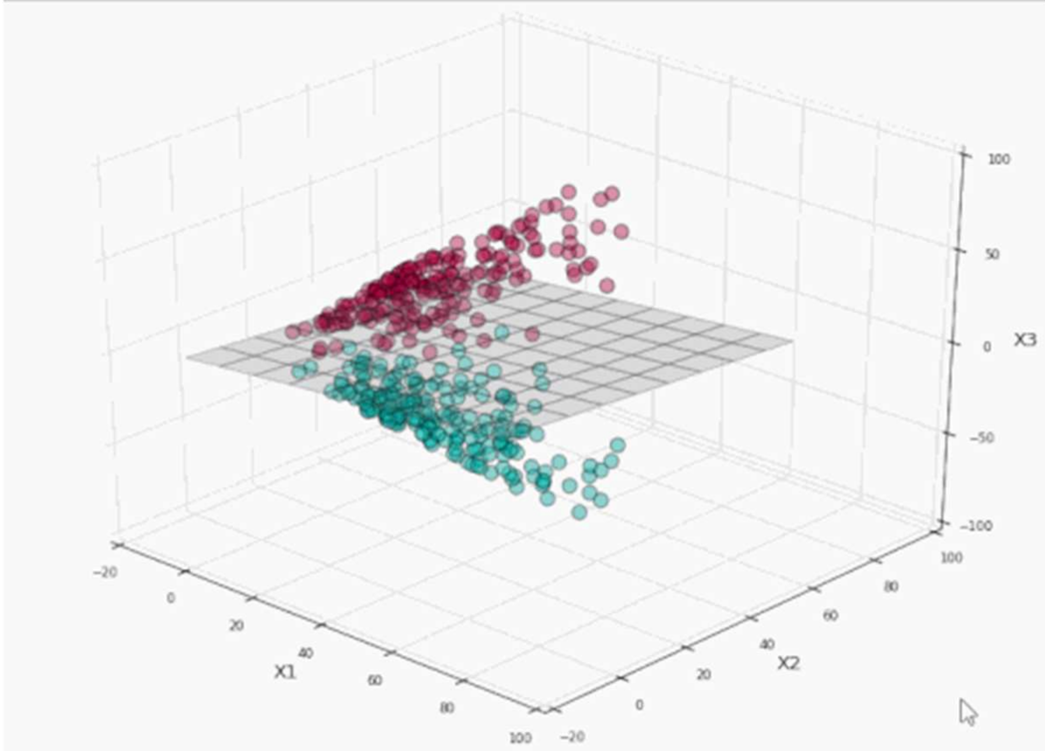


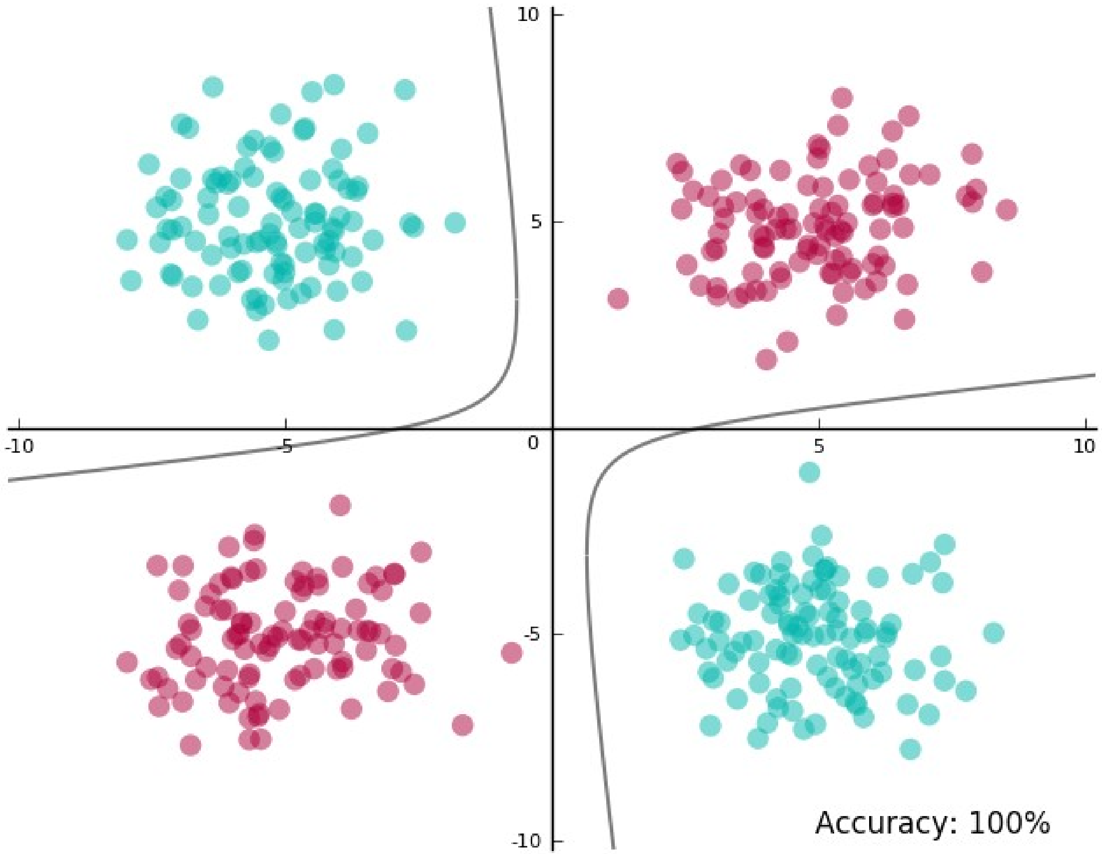

C=0.01
C=0.10
C=1.00
C=10.00

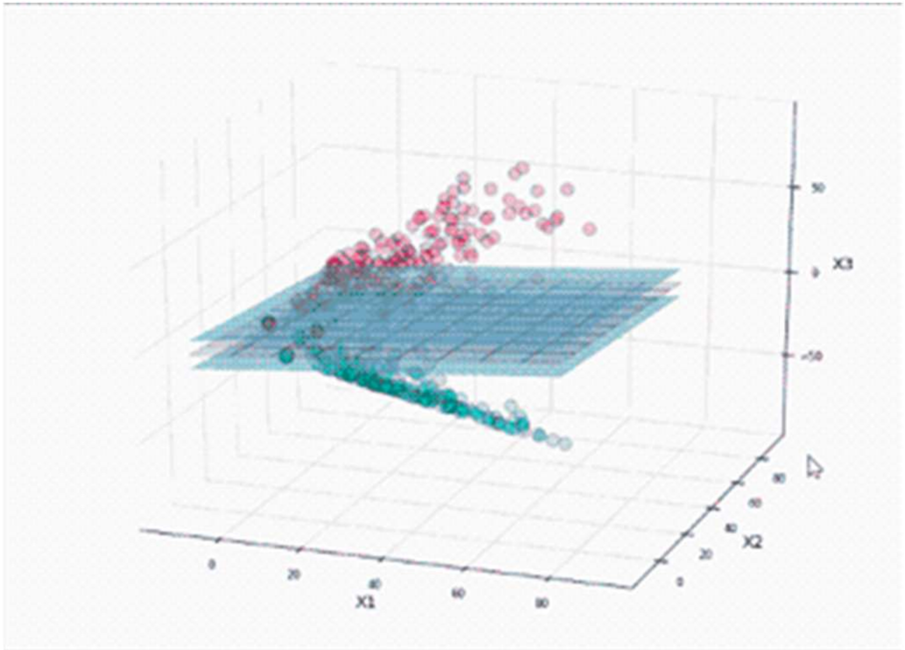
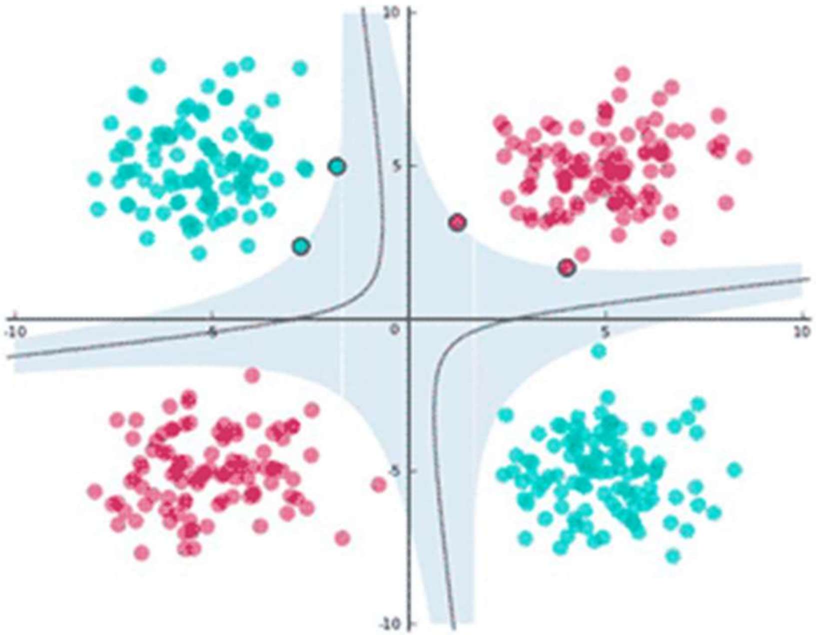


10
5
-10
-5
0
5
10
-5
-10
Accuracy: 75%


X1
X2
X3


X1
X2
X3

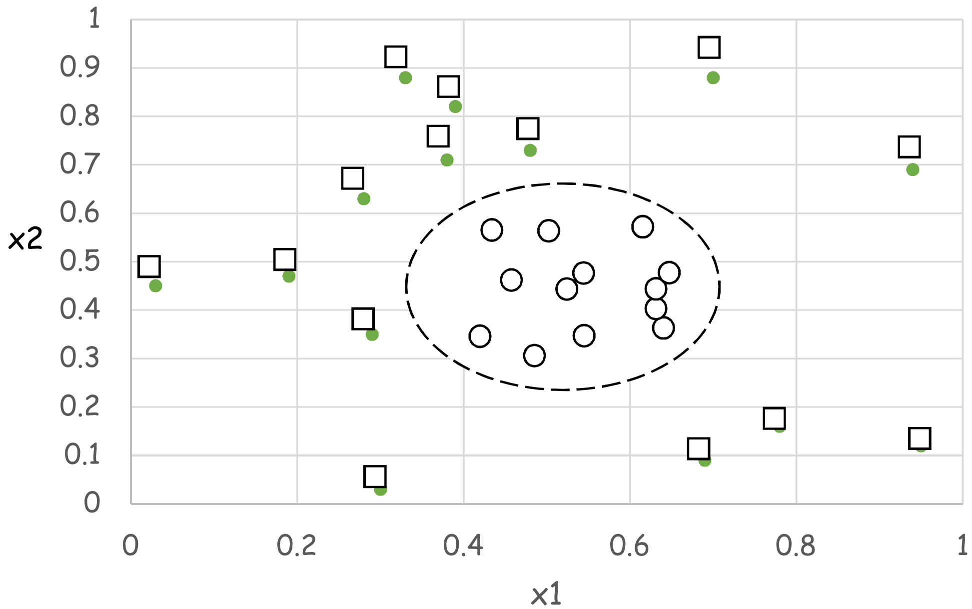

10
5
-10
-5
0
5
10
-5
-10
Accuracy: 100%


X1
X2
X3


非线性支持向量机

- 数据集:所有圆圈部分都聚集在图的中心附近，而所有的方块都分布在离中心较远的地方。
- 方块（类标号**y=1**）和圆圈（类标号**y=-1**）。可以使用下面的公式对数据集中的实例分类。

$$y(x_1,x_2)=\begin{cases}1, & if\sqrt{(x_1-0.5)^2+(x_2-0.5)^2}>0.2\\ -1, & otherwise\end{cases}$$



原二维空间中的决策边界

非线性支持向量机

因此，数据集的决策边界可以表示如下：

$$\sqrt{(x_1-0.5)^2+(x_2-0.5)^2}=0.2$$

进一步简化为下面的二次方程：

$$x_1^2-x_1+x_2^2-x_2=-0.46$$

需要一个非线性变换Φ，将数据从原先的特征空间映射到新的空间，决策边界在这个空间下成为线性的，假定选择下面的变换：

$$\Phi:(x_1,x_2)\rightarrow(x_1^2,x_2^2,\sqrt{2}x_1,\sqrt{2}x_2,\sqrt{2}x_1x_2,1)$$

在变换后的空间，选择参数：

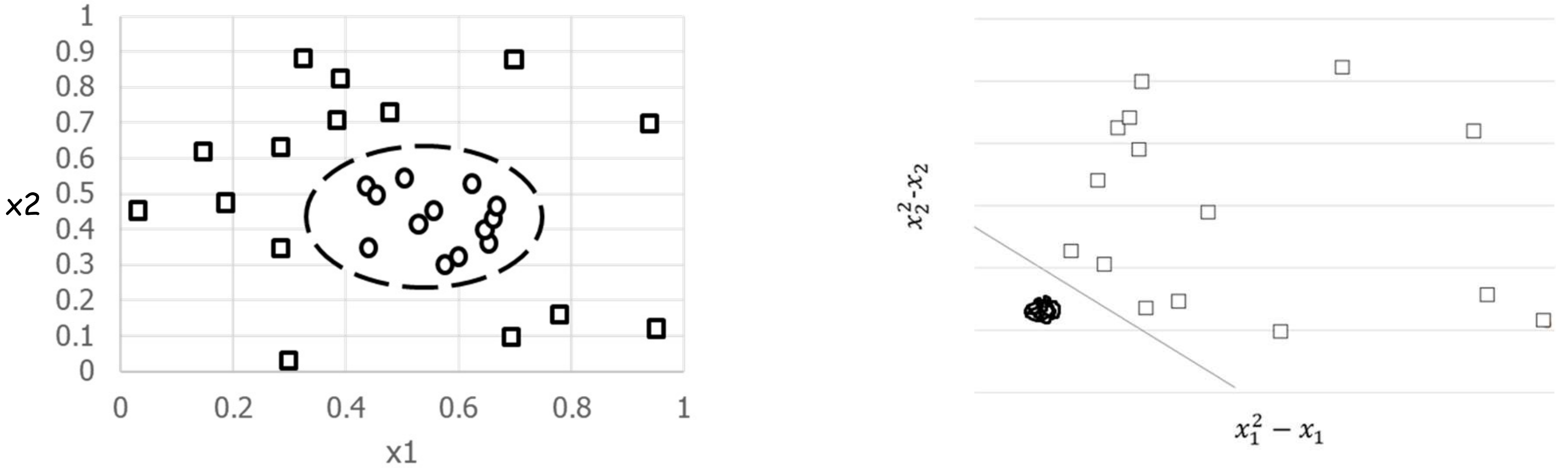
$$w=(w_0,w_1,\ldots,w_5)$$

使得：

$$w_5x_1^2+w_4x_2^2+w_3\sqrt{2}x_1+w_2\sqrt{2}x_2+w_1\sqrt{2}x_1x_2+w_0=0$$

非线性支持向量机

- 例，对于图(a)给定的数据，以$x_1^2-x_1$和$x_2^2-x_2$为坐标绘图。图(b) 显示在变换后的空间中，所有的圆圈都位于图的左下方，可构建一个线性的决策边界从而把数据划分到各自所属的类中。



（a）原二维空间中的决策边界　　（b）变换后空间的决策边界

图　分类具有非线性决策边界的数据

核函数：一种使用原属性集计算变化后的空间的相似度的方法。

$\Phi(x_i)\cdot\Phi(x_j)$可看作两个实例x_i、x_j在变化后的空间中的相似性度量

$$\Phi:(x_1,x_2)\rightarrow(x_1^2,x_2^2,\sqrt{2}x_1,\sqrt{2}x_2,\sqrt{2}x_1x_2,1)$$

两个输入向量u和v在变化后的空间中的点积可写成：

$$\begin{aligned}\Phi(u)\cdot\Phi(v) &= (u_1^2,u_2^2,\sqrt{2}u_1,\sqrt{2}u_2,\sqrt{2}u_1u_2,1)\cdot(v_1^2,v_2^2,\sqrt{2}v_1,\sqrt{2}v_2,\sqrt{2}v_1v_2,1)\\ &= u_1^2v_1^2+u_2v_2^2+2u_1v_1+2u_2v_2+2u_1u_2v_1v_2+1\\ &= (uv+1)^2\end{aligned}$$

变换后的空间中的点积可以用原空间中的相似度函数表示：

$$K(u,v)=\Phi(u)\cdot\Phi(v)=(uv+1)^2$$

这个在原属性空间中计算的相似度函数K称为核函数。

检验实例z可以通过下式分类：

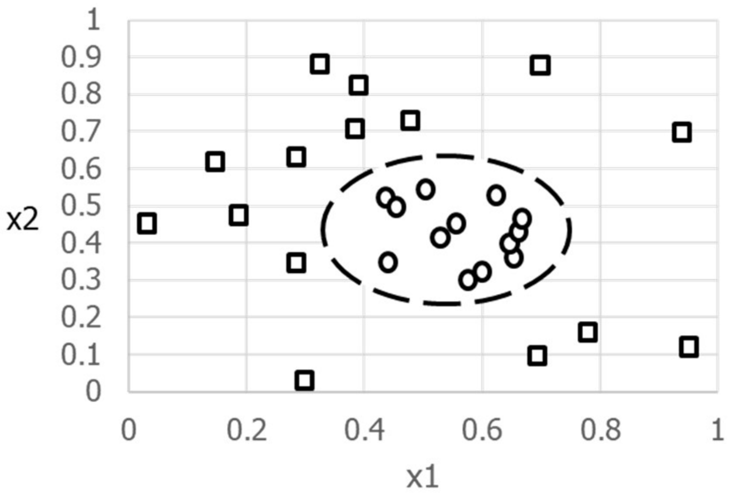
$$K(u,v)=\Phi(u)\Phi(v)=(uv+1)^2$$

$$f(z) \quad = sign(\textstyle\sum_{i=1}^{n}\alpha_i\, y_i\Phi(x_i)\cdot\Phi(z)+b)$$

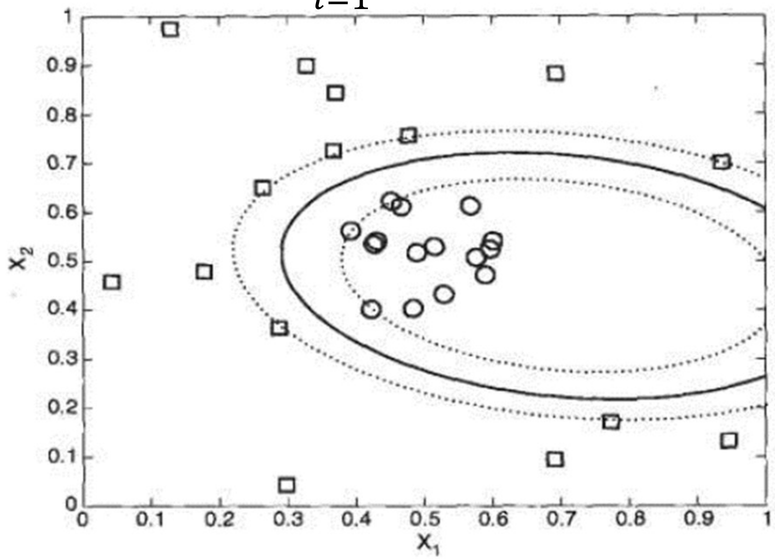
$$K(u,v) = \Phi(u) \cdot \Phi(v) = (uv+1)^2$$

检验实例z可以通过下式分类：

$$\begin{aligned} f(z) &= sign(\sum_{i=1}^{n} \lambda_i \, y_i \Phi(x_i) \cdot \Phi(z) + b) \\ &= sign(\sum_{i=1}^{n} \lambda_i \, y_i K(x_i, z) + b) \\ &= sign(\sum_{i=1}^{n} \lambda_i \, y_i (x_i \cdot z + 1)^2 + b) \end{aligned}$$



原二维空间中的决策边界



具有多项式的非线性SVM产生的决策边界

非线性支持向量机

- 核函数：将数据点转换到另一个高维特征空间(Christopher J.C Burges,1998)，在这个空间中这些数据点是可以被线性分割的。
- 设这个转换为$\Phi(.)$，在高维空间中，解决下面的问题

$$\Psi_D = \sum_{i=1}^{N} \alpha_i - \sum_{i=1}^{N} \frac{1}{2} \alpha_i \alpha_j y_i y_j \Phi(x_i) \cdot \Phi(x_j)$$

- 假设$\Phi(x_i) \cdot \Phi(x_j) = k(x_i, x_j)$，即在高维空间中点积相当于输入空间的一个核函数。
- 常用核函数
  - 简单的点积（simple dot product）

$$K(x, y) = x \cdot y$$

  - 多项式核函数（polynomial function）

$$K(x, y) = \left((x \cdot y) + 1\right)^d$$

非线性支持向量机

- **Vovk**实数多项式核函数：

$$K(x,y)=\frac{1-(x\cdot y)^{d}}{1-(x\cdot y)}$$

- **Vovk**实数无穷大多项式核函数：

$$K(x,y)=\frac{1}{1-(x\cdot y)} \quad -1<(x\cdot y)<1$$

- 高斯径向基核函数（**Gaussian RBF**）　　**δ**是参数

$$K(x_a,x_b)=\exp\left(-\frac{\|x_a-x_b\|^2}{2\delta^2}\right)$$

得到的分割超平面：

$$f(x)=\mathrm{sign}\left[\sum_i \alpha_i y_i \exp\left(-\frac{\|x-x_i\|^2}{2\delta^2}\right)+b\right]$$

非线性支持向量机

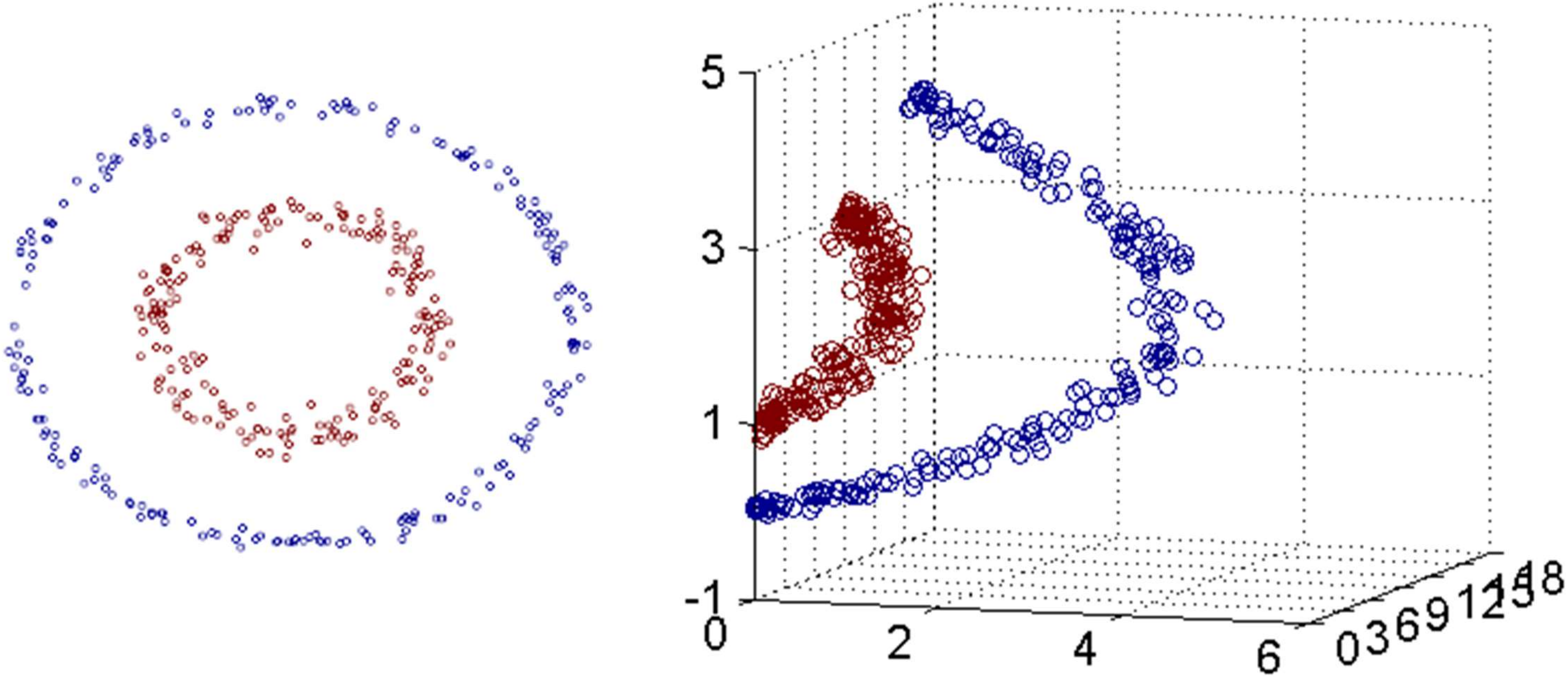
- 两层神经网络核函数（sigmoid）

$$\tanh\left(\frac{b(xy)}{1} - c\right)$$

b和c是用户自己定义

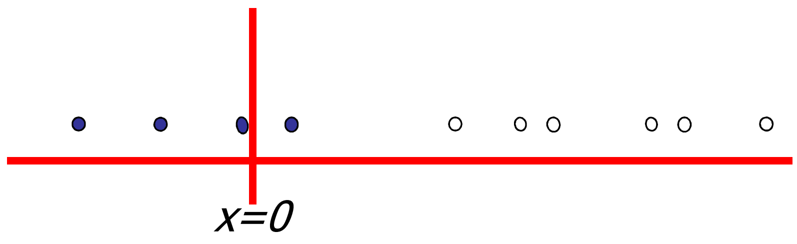
- Kernel Tricks
  - Replacing dot product with a kernel function
  - Not all functions are kernel functions
    - Need to be decomposable
      - $K(a,b) = \overrightarrow{\phi(a)} \cdot \overrightarrow{\phi(b)}$
    - Could $K(a,b) = (a-b)^3$ be a kernel function ?
    - Could $K(a,b) = (a-b)^4 - (a+b)^2$ be a kernel function?
- 例输入空间是R^2， 核函数$K(x,z)=(x\cdot z)^2$，试找出其相关的新空间和映射$\Phi(x):R^2$->新空间
  - 新空间R3

  - 新空间R4

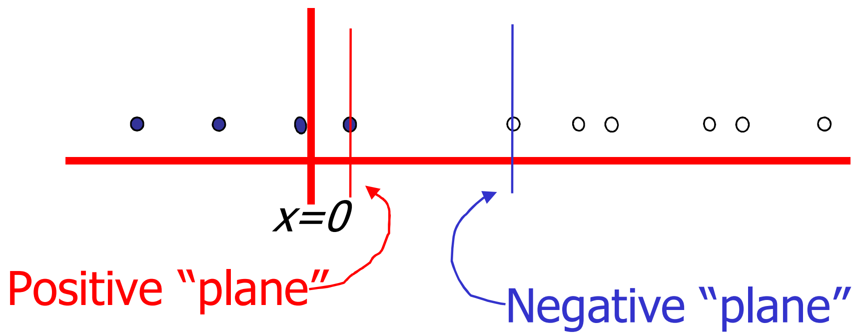

5
3
1
-1
0
2
4
6
0
3
6
9
12

Suppose we're in 1-dimension

What would SVMs do with this data?

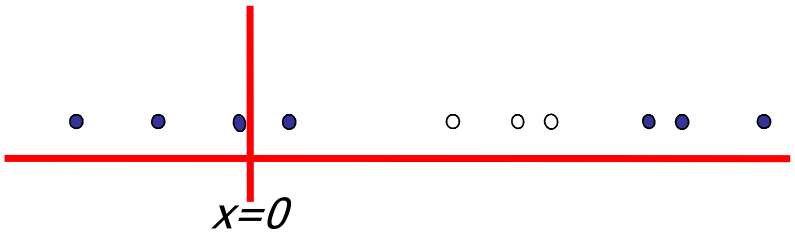


Suppose we're in 1-dimension



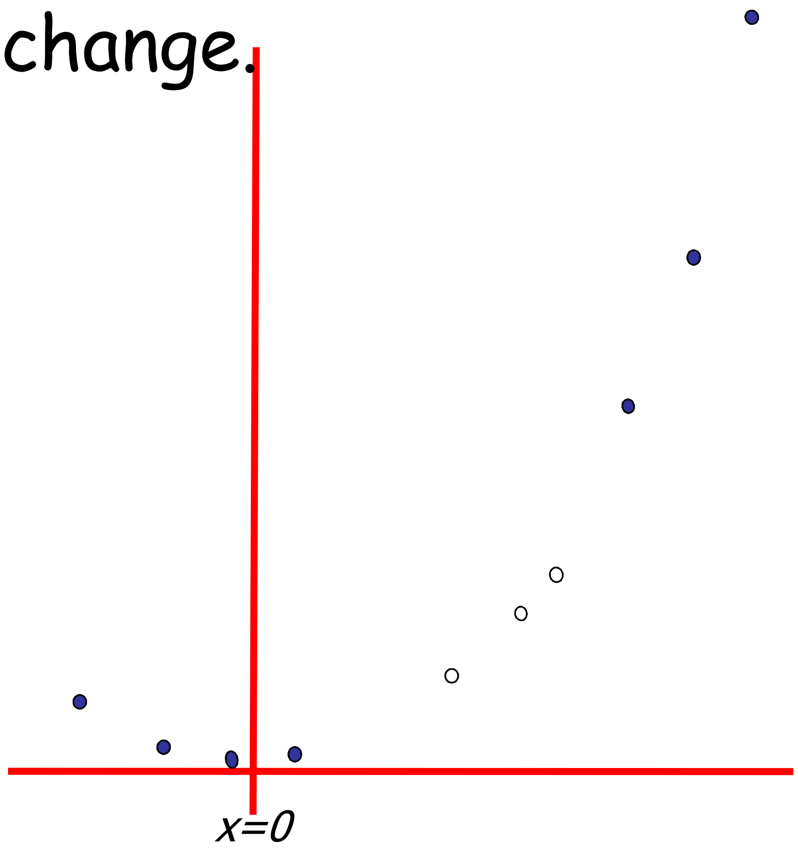
Harder 1-dimensional dataset

What can be done about this?



Harder 1-dimensional dataset

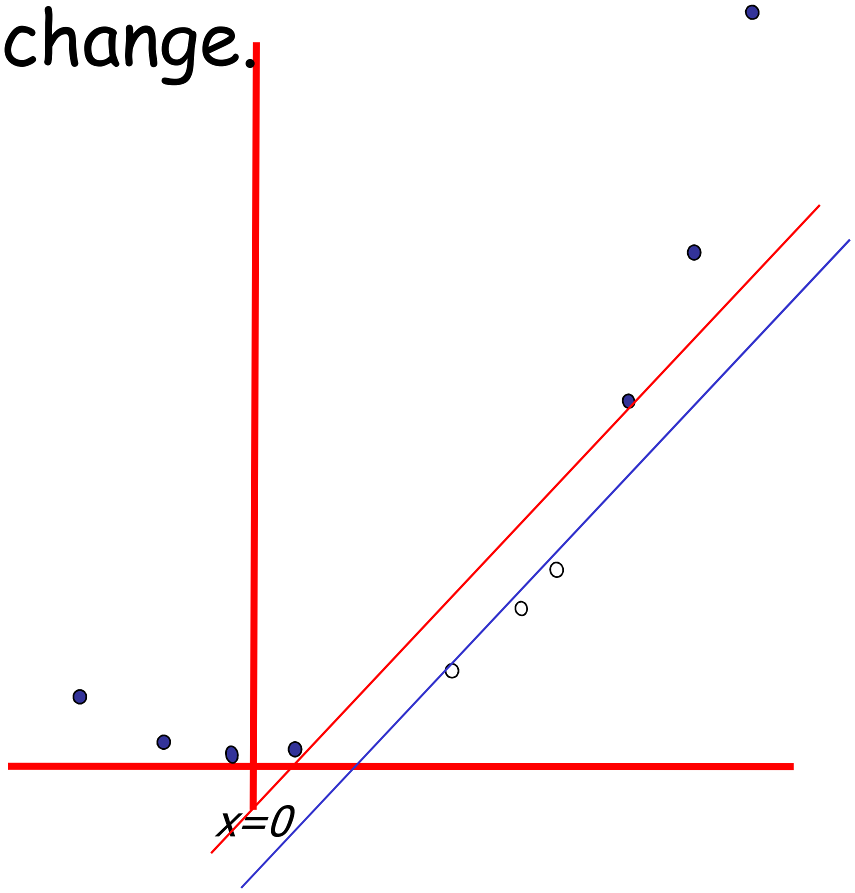
Let's do some change.



$$\mathbf{z}_k = (x_k, x_k^2)$$

Harder 1-dimensional dataset

Let's do some change.



$$\mathbf{z}_k = (x_k, x_k^2)$$

An Example

- Suppose we have 5 1D data points
  - $x_1=1$, $x_2=2$, $x_3=4$, $x_4=5$, $x_5=6$, with 1, 2, 5 as class 1 and 3, 4 as class 2 $\Rightarrow$ $y_1=1$, $y_2=1$, $y_3=-1$, $y_4=-1$, $y_5=1$
- We use the polynomial kernel of degree 2
  - $K(x,z) = (xz+1)^2$
  - C is set to 100
- We first find α_i (i=1, …, 5) by

$$\text{max.} \quad \sum_{i=1}^{5} \alpha_i - \frac{1}{2} \sum_{i=1}^{5} \sum_{j=1}^{5} \alpha_i \alpha_j y_i y_j (x_i x_j + 1)^2$$

$$\text{subject to } 100 \geq \alpha_i \geq 0, \sum_{i=1}^{5} \alpha_i y_i = 0$$

An Example

- By using the method, we get

  $\forall \alpha_1=0,\ \alpha_2=2.5,\ \alpha_3=0,\ \alpha_4=7.333,\ \alpha_5=4.833$

  - Note that the constraints are indeed satisfied
  - The support vectors are $\{x_2=2,\ x_4=5,\ x_5=6\}$

- The discriminant function is

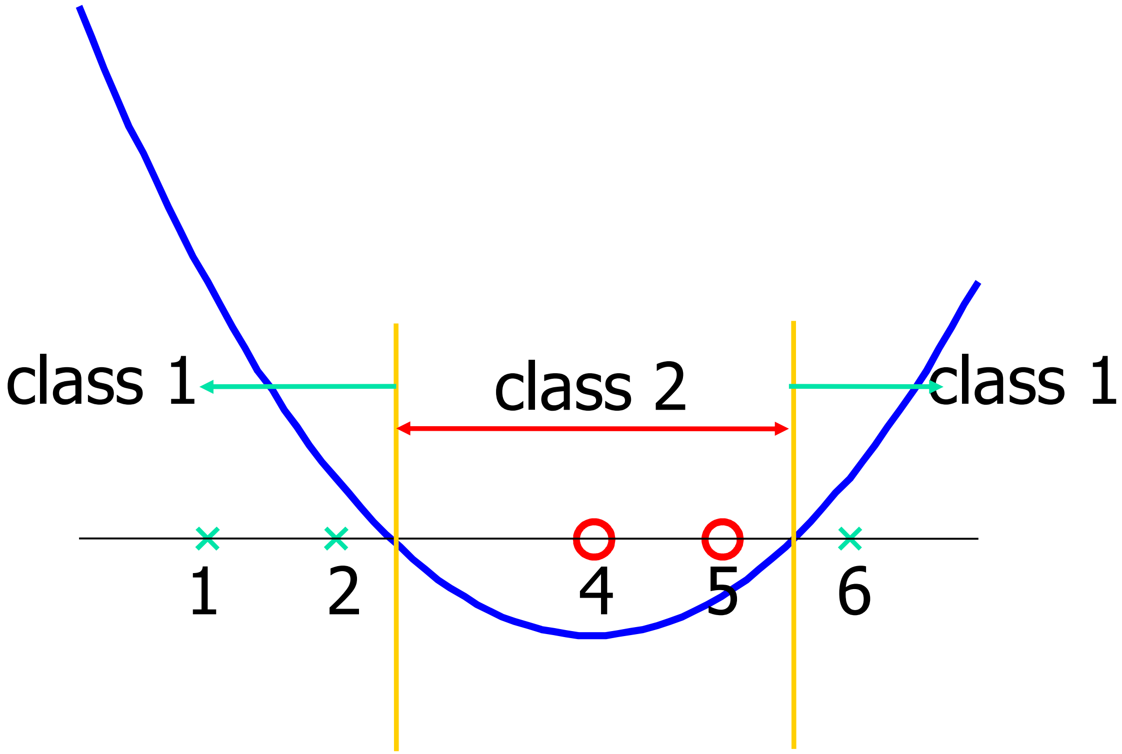
$$
\begin{aligned}
&f(z) \\
&= 2.5(1)(2z+1)^2 + 7.333(-1)(5z+1)^2 + \overset{\alpha_5}{4.833}(\overset{y_5}{1})\overset{K(z,x_5)}{(6z+1)^2} + b \\
&= 0.6667z^2 - 5.333z + b
\end{aligned}
$$

- b is recovered by solving $f(2)=1$ or by $f(5)=-1$ or by $f(6)=1$, as x_2 and x_5 lie on the line $\phi(\mathbf{w})^T\phi(\mathbf{x})+b=1$ and x_4 lies on the line $\phi(\mathbf{w})^T\phi(\mathbf{x})+b=-1$

- All three give $b=9$ $\Longrightarrow$ $f(z) = 0.6667z^2 - 5.333z + 9$

Another Example

Value of discriminant function



Summary: Steps for Classification

- Prepare the pattern matrix
- Select the kernel function to use
- Select the parameter of the kernel function and the value of *C*
  - You can use the values suggested by the SVM software, or you can set apart a validation set to determine the values of the parameter
- Execute the training algorithm and obtain the α_i
- Unseen data can be classified using the α_i and the support vectors

转导式支持向量机

- 转导式学习在训练过程中使用较少的有标签样本和较多的无标签样本进行学习。
- 测试集的样本分布信息从无标签样本转移到了最终的分类器中。
- 由于无标签样本的数量较多，所以与有标签样本相比更能够很好地刻画整个样本空间上的数据特性，从而使训练出的分类器具有更好的推广性能。

多分类支持向量机

- One-against-rest
  - 构建了k个SVM模型，k是类别的个数。
  - 第i个SVM的：属于第i个类别的实例具有正例标记，属于其它所有类别的实例具有反例标记。
  - 对于给定的l个训练数据$(x_1, y_1), (x_2, y_2), \dots, (x_l, y_l)$,则第i个SVM解决下面的问题：
    - $\underset{w^i, b^i, \xi^i}{\text{Min}} \ \frac{1}{2}\left(w^i\right)^T w^i + C\sum_{j=1}^{l} \xi_j^i$

$$\left(w^i\right)^T \Phi\left(x_j\right) + b^i \geq 1 - \xi_j^i, \text{ if } y_j = i$$

$$\left(w^i\right)^T \Phi\left(x_j\right) + b^i \leq -1 + \xi_j^i, \text{ if } y_j \neq i$$

$$\xi_j^i \geq 0, j = 1, \cdots, l$$

多分类支持向量机

- 得到**k**个决策函数

$$(w^1)^T\Phi(x)+b^1$$
$$\vdots$$
$$\left(w^k\right)^T\Phi(x)+b^k$$

- ***x***属于最大决策函数值所对应的类别：

$$\text{class of } x=\arg\max_{i=i,\cdots,k}\left(\left(w^i\right)^T\Phi(x)+b^i\right)$$

多分类支持向量机

- Pairwise
  - 构建 $k(k-1)/2$ 个二分类器，来自两个类别的数据训练一个分类器，即，对于来自第i类和第j类的训练数据，解决下面的两类别分类问题：

$$\min_{w^i,b^i,\xi^i} \frac{1}{2}\left(w^{ij}\right)^T w^{ij} + C\sum_{j=1}^{l} \xi_j^{ij}$$

$$\left(w^{ij}\right)^T \Phi\left(x_j\right) + b^{ij} \geq 1 - \xi_j^{ij}, \quad \text{if } y_j = i$$

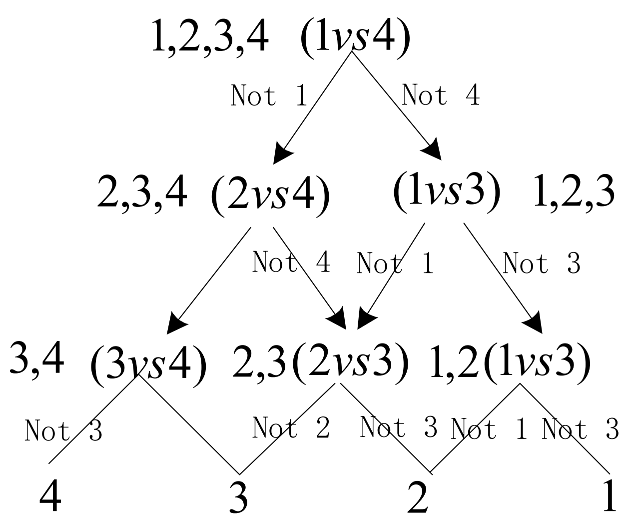
$$\left(w^{ij}\right)^T \Phi\left(x_j\right) + b^{ij} \leq -1 + \xi_j^{ij}, \quad \text{if } y_j \neq i$$

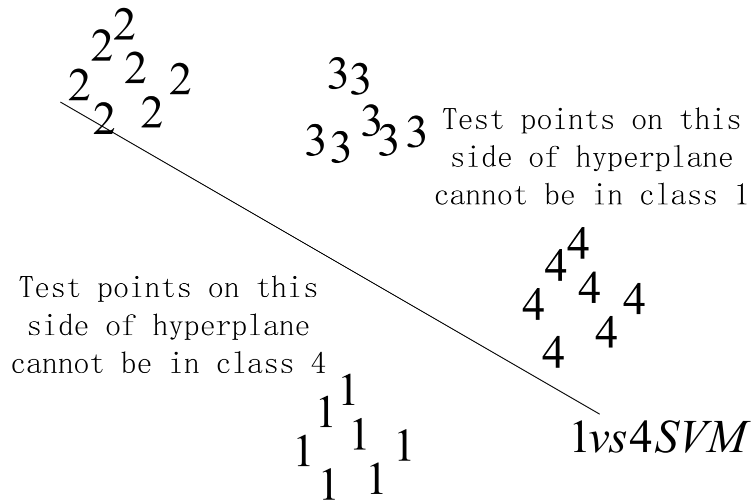
$$\xi_t^{ij} \geq 0$$

多分类支持向量机

- $k(k-1)/2$个分类器构建完后，使用选举策略(J.Friedman 1996)
- 如果$\text{sign}\left(\left(w^{ij}\right)^T\Phi(x)+b^{ij}\right)$判断样本***x***在第*i*类中，于是对第*i*类的选票加一，否则对第*j*类的选票加一；
- 根据投票的结果，预测出***x***属于投票最多的类别。

- DAGSVM
  - 有向的非循环图支持向量机(DAGSVM)(John.C.Platt,2000)。训练阶段与pairwise方法是一样的，构建$k(k-1)/2$个两类别的SVMs，其中每一个SVMs解决两个类别的分类问题。
  - 测试阶段，使用具有$k(k-1)/2$个内部结点和k个叶子结点的一个关于根结点的两类别有向非循环图来进行分类。





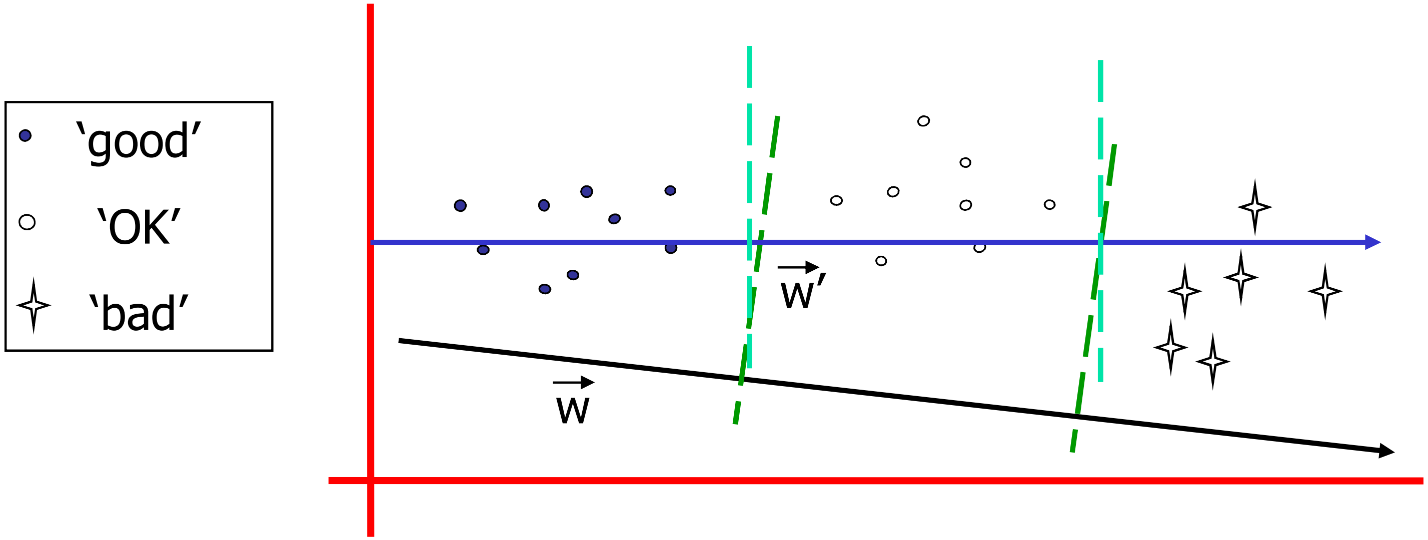
SVM Performance

- Anecdotally they work very well indeed.
- Example: They are currently the best-known classifier on a well-studied hand-written-character recognition benchmark
- Several reliable people doing practical real-world work who claim that SVMs have saved them when their other favorite classifiers did poorly.

Ranking Problem

- Consider a problem of ranking essays
  - Three ranking categories: good, ok, bad
  - Given a input document, predict its ranking category
- How should we formulate this problem?
- A simple multiple class solution
  - Each ranking category is a independent class

Ordinal Regression



- Which choice is better?
- How could we formulate this problem?

Ordinal Regression

- What are the two decision boundaries?

$$\mathbf{w} \cdot \mathbf{x} + b_1 = 0 \text{ and } \mathbf{w} \cdot \mathbf{x} + b_2 = 0$$

- What is the margin for ordinal regression?

$$\text{margin}_1(\mathbf{w}, b_1) \equiv \min_{\mathbf{x} \in D_g \cup D_o} d(\mathbf{x}) = \min_{\mathbf{x} \in D_g \cup D_o} \frac{\left|\mathbf{x} \cdot \mathbf{w} + b_1\right|}{\sqrt{\sum_{i=1}^{d} w_i^2}}$$

$$\text{margin}_2(\mathbf{w}, b_2) \equiv \min_{\mathbf{x} \in D_o \cup D_b} d(\mathbf{x}) = \min_{\mathbf{x} \in D_o \cup D_b} \frac{\left|\mathbf{x} \cdot \mathbf{w} + b_2\right|}{\sqrt{\sum_{i=1}^{d} w_i^2}}$$

$$\text{margin}(\mathbf{w}, b_1, b_2) = \min(\text{margin}_1(\mathbf{w}, b_1), \text{margin}_2(\mathbf{w}, b_2))$$

- Maximize margin $\{\mathbf{w}^*, b_1^*, b_2^*\} = \arg\max_{\mathbf{w}, b_1, b_2} \text{margin}(\mathbf{w}, b_1, b_2)$

Ordinal Regression

$$
\begin{aligned}
\{\mathbf{w}^*, b_1^*, b_2^*\} &= \arg\max_{\mathbf{w}, b_1, b_2} \ \text{margin}(\mathbf{w}, b_1, b_2) \\
&= \arg\max_{\mathbf{w}, b_1, b_2} \ \min(\text{margin}_1(\mathbf{w}, b_1), \text{margin}_2(\mathbf{w}, b_2)) \\
&= \arg\max_{\mathbf{w}, b_1, b_2} \ \min\left(\min_{\mathbf{x} \in D_g \cup D_o} \frac{|\mathbf{x} \cdot \mathbf{w} + b_1|}{\sqrt{\sum_{i=1}^{d} w_i^2}}, \min_{\mathbf{x} \in D_o \cup D_b} \frac{|\mathbf{x} \cdot \mathbf{w} + b_2|}{\sqrt{\sum_{i=1}^{d} w_i^2}} \right)
\end{aligned}
$$

subject to

$$
\begin{aligned}
&\forall \mathbf{x}_i \in D_g : \mathbf{x}_i \cdot \mathbf{w} + b_1 > 0 \\
&\forall \mathbf{x}_i \in D_o : \mathbf{x}_i \cdot \mathbf{w} + b_1 < 0, \mathbf{x}_i \cdot \mathbf{w} + b_2 > 0 \\
&\forall \mathbf{x}_i \in D_b : \mathbf{x}_i \cdot \mathbf{w} + b_2 < 0
\end{aligned}
$$

- How do we solve it?

Ordinal Regression

- The same old trick
- To remove the scaling invariance, set

$$\forall \mathbf{x}_i \in D_g \cup D_o : \ \left| b_1 + \mathbf{x}_i \cdot \mathbf{w} \right| \geq 1$$

$$\forall \mathbf{x}_i \in D_o \cup D_b : \ \left| b_2 + \mathbf{x}_i \cdot \mathbf{w} \right| \geq 1$$

- Now the problem is simplified as:

$$\{\mathbf{w}^*, b_1^*, b_2^*\} = \underset{\mathbf{w}, b_1, b_2}{\arg\min} \ \sum_{i=1}^{d} w_i^2$$

subject to

$$\forall \mathbf{x}_i \in D_g : \mathbf{x}_i \cdot \mathbf{w} + b_1 \geq 1$$

$$\forall \mathbf{x}_i \in D_o : \mathbf{x}_i \cdot \mathbf{w} + b_1 \leq -1, \mathbf{x}_i \cdot \mathbf{w} + b_2 \geq 1$$

$$\forall \mathbf{x}_i \in D_b : \mathbf{x}_i \cdot \mathbf{w} + b_2 \leq -1$$

Ordinal Regression

- Noisy case

$$\{\mathbf{w}^*, b_1^*, b_2^*\} = \underset{\mathbf{w}, b_1, b_2}{\arg\min} \sum_{i=1}^d w_i^2 + c \sum_{x_i \in D_g} \varepsilon_i^+ + c \sum_{x_i \in D_o} \left(\varepsilon_i^+ + \varepsilon_i^-\right) + c \sum_{x_i \in D_b} \varepsilon_i^-$$

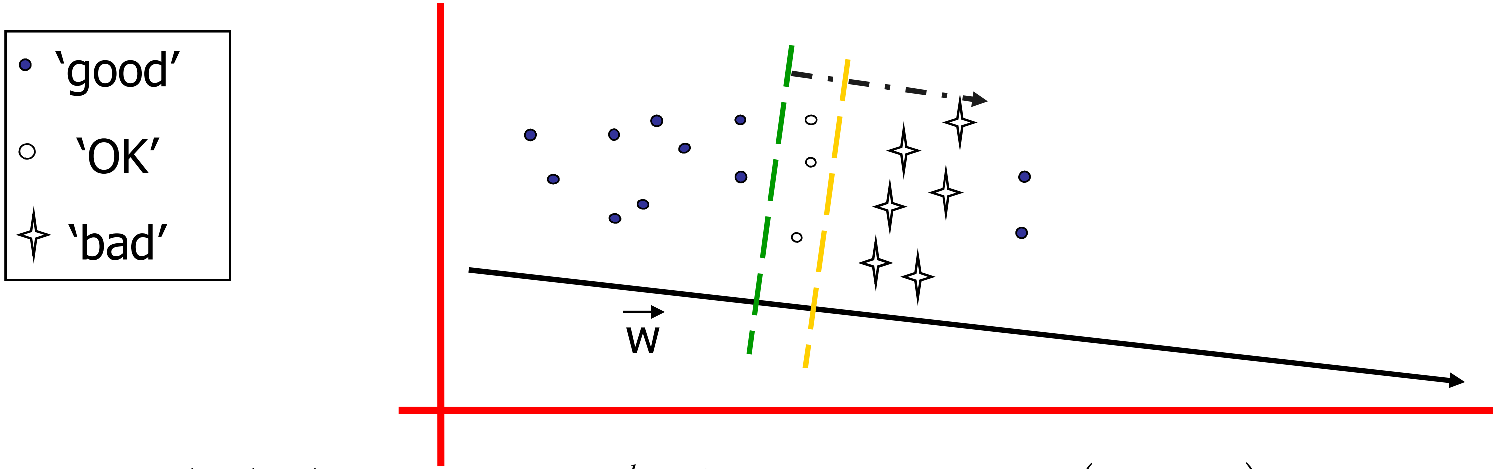
subject to

$$\forall \mathbf{x}_i \in D_g : \mathbf{x}_i \cdot \mathbf{w} + b_1 \geq 1 - \varepsilon_i^+, \varepsilon_i^+ \geq 0$$

$$\forall \mathbf{x}_i \in D_o : \mathbf{x}_i \cdot \mathbf{w} + b_1 \leq -1 + \varepsilon_i^-, \mathbf{x}_i \cdot \mathbf{w} + b_2 \geq 1 - \varepsilon_i^+, \varepsilon_i^+ \geq 0, \varepsilon_i^- \geq 0$$

$$\forall \mathbf{x}_i \in D_b : \mathbf{x}_i \cdot \mathbf{w} + b_2 \leq -1 + \varepsilon_i^-, \varepsilon_i^- \geq 0$$

Ordinal Regression



$$\{\mathbf{w}^*, b_1^*, b_2^*\} = \underset{\mathbf{w}, b_1, b_2}{\arg\min} \sum_{i=1}^d w_i^2 + c \sum_{x_i \in D_g} \varepsilon_i^+ + c \sum_{x_i \in D_o} \left(\varepsilon_i^+ + \varepsilon_i^-\right) + c \sum_{x_i \in D_b} \varepsilon_i^-$$

subject to

$$\forall \mathbf{x}_i \in D_g : \mathbf{x}_i \cdot \mathbf{w} + b_1 \geq 1 - \varepsilon_i^+, \varepsilon_i^+ \geq 0$$

$$\forall \mathbf{x}_i \in D_o : \mathbf{x}_i \cdot \mathbf{w} + b_1 \leq -1 + \varepsilon_i^-, \mathbf{x}_i \cdot \mathbf{w} + b_2 \geq 1 - \varepsilon_i^+, \varepsilon_i^+ \geq 0, \varepsilon_i^- \geq 0$$

$$\forall \mathbf{x}_i \in D_b : \mathbf{x}_i \cdot \mathbf{w} + b_2 \leq -1 + \varepsilon_i^-, \varepsilon_i^- \geq 0$$

$$\boxed{b_1 \leq b_2}$$

Resources

- http://www.cs.cmu.edu/~awm/tutorials
- http://www.kernel-machines.org/
- http://www.support-vector.net/
- http://www.support-vector.net/icml-tutorial.pdf
- http://www.kernel-machines.org/papers/tutorial-nips.ps.gz
- http://www.clopinet.com/isabelle/Projects/SVM/applist.html

- http://www.csie.ntu.edu.tw/~cjlin/libsvm/
- http://www.csie.ntu.edu.tw/~cjlin/liblinear/
- http://svmlight.joachims.org/svm_struct.html
- http://www.cs.huji.ac.il/~shais/code/index.html

Homework

1.2 已知正例点 $x_1=(1,2)^{\mathrm{T}}$，$x_2=(2,3)^{\mathrm{T}}$，$x_3=(3,3)^{\mathrm{T}}$，负例点 $x_4=(2,1)^{\mathrm{T}}$，$x_5=(3,2)^{\mathrm{T}}$，试求最大间隔分离超平面和分类决策函数，并在图上画出分离超平面、间隔边界及支持向量.

LIBSVM 见 http://www.csie.ntu.edu.tw/~cjlin/libsvm/.

UCI 数据集见 http://archive.ics.uci.edu/ml/.

6.3 选择两个 UCI 数据集，分别用线性核和高斯核训练一个 SVM，并与 ~~BP 神经网络~~和 C4.5 决策树进行实验比较.

- [Python 基础教程 (runoob.com)]

Thank You!